Jingkun Jiang

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Education

Hunan University, China Sep 2024 - Jun 2027
M.S. in Materials Science and Engineering Advisor: Prof. **Yang Xia**

- Scientific Computing, Phase-Field Modeling, High-Performance Computing (HPC)

Hebei University of Architecture, China Sep 2020 - Jun 2024
B.Eng. in Materials Science and Engineering **Rank 3/73 (5%)**

Publications & Patents

[1] **J. Jiang**, P. Deng, and Y. Xia. "sp-DBA: a general framework for adaptive transform-domain computation." Under review at *Communications AI & Computing*, 2026. [**arXiv**]

[P1] **J. Jiang** and Y. Xia. "A GPU-Oriented Adaptive Acceleration Method Using Spectral Dynamic Block Activation in the Fourier Domain." Chinese Invention Patent Application, No. 202611350264.9, 2026.

Research Experience

sp-DBA Framework for Adaptive Fourier-Domain Computation — Sep 2024 - Present

- Led the design and implementation of **sp-DBA**, a **GPU-friendly adaptive computing framework** for Fourier-domain computation with hardware-aligned block activation and numerical stability controls. Achieved up to **28.1× speedup** in the main update step and **8.4× overall speedup** while preserving accuracy.
- Applied sp-DBA to **phase-field crystal (PFC)** simulations for polycrystalline coarsening, multicomponent **reaction-diffusion** models for biological pattern formation, and **optical signal processing**.
- Designed sp-DBA with a shared core and lightweight solver interfaces, adding about **8-16% additional solver code** while retaining the existing governing equations and numerical schemes.

Computational Materials Modeling with Phase-Field Methods — Sep 2024 - Present

- Developed a 3D **phase-field (PF)** solver for dendritic growth, and **PFC, XPFC, and APFC** solvers for **crystalline microstructure evolution** including grain coarsening, interface growth, and defect evolution.
- Implemented finite-difference and Fourier pseudo-spectral solvers in **C++**, using fourth-order exponential time-differencing (ETD4) schemes for PFC/XPFC simulations, and accelerated large-scale simulations using **OpenMP and MPI parallelization**.
- Reconstructed grains from PFC density fields and quantified **grain-size distributions (GSD) and coarsening behavior** in large-scale microstructure analysis.

Distributed FFT and Multi-GPU Scaling — Dec 2025 - Present

- Built a distributed 2D FFT using slab decomposition, local cuFFT transforms, MPI-based redistribution, and CUDA-aware communication to support distributed sp-DBA.
- Evaluated **strong and weak scaling on 1-8 NVIDIA A800 GPUs** and identified communication overhead in distributed FFT as the main performance bottleneck at larger GPU counts.

Skills

Programming	C/C++, Python, MATLAB
Parallel/HPC	CUDA, OpenMP, MPI, distributed FFTs, multi-GPU scaling
Software	ParaView (PF/PFC), OVITO, VASP (DFT), Materials Studio

Honors & Academic Service

Provincial First Prize, Chinese Mathematics Competitions for College Students	2022, 2023
Outstanding Graduate, Hebei University of Architecture	2024
Teaching Assistant, Hunan University	2025 - 2026
Student Affairs Assistant, Hunan University	2026 - Present